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|------|-------------------------------------------------------------------------------------------------|---|
| 2(a) | cholera = prokaryotic<br><u>and</u><br>malaria = eukaryotic<br><u>and</u><br>TB = prokaryotic ; | 1 |
| 2(b) | pathogen ;                                                                                      | 1 |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
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| 2(c)(i)  | <p>magnification (x) 12 000 ;; for line X-Y measuring 24 mm<br/> <b>A</b> calculated values for line X-Y measured between 23 to 25 mm</p> <p>for one mark:</p> <ul style="list-style-type: none"> <li>• correct calculation for 23.5 or 24.5 but not to 3 sig. figs = 1 mark</li> <li>• incorrect answer e.g. conversion factor incorrect, but <ul style="list-style-type: none"> <li>– shows correct measurement for image</li> <li>– uses correct formula for calculation</li> <li>– answer to 3 sig. figs</li> </ul> </li> <li>• measurement 22 mm or 26 mm <ul style="list-style-type: none"> <li>– correct answer for their measurement</li> <li>– uses correct formula for calculation</li> <li>– answer to 3 sig. figs</li> </ul> </li> </ul> | 2     |
| 2(c)(ii) | <p>any <b>three</b> from:</p> <p><b>1</b> by a vector / carrier / mosquito / insect / (female) <i>Anopheles</i> ;</p> <p><b>2</b> takes, blood meal / sucks blood / feeds on blood ;<br/> context is transmission from infected to uninfected</p> <p><b>3</b> <i>Plasmodium</i> / pathogen / parasite, is in blood ; in correct context</p> <p><b>4</b> injects <i>Plasmodium</i> / AW, together with, anticoagulant / saliva ;</p> <p><b>5</b> AVP ; e.g. <i>Plasmodium</i> / AW, migrates, after ingestion / from insect gut / AW, to salivary glands</p> <p>if another mode of transmission stated, e.g. transfusion, allow mp1 and mp3 and a relevant AVP</p>                                                                                    | 3     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
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| 2(d)(i)  | <p>any <b>two</b> from:</p> <p>(phagocyte) has (specific) <u>receptors</u> (for PS) ;</p> <p><b>R</b> antigens</p> <p><b>A</b> named e.g. macrophage / monocyte / Kupffer cell / neutrophil</p> <p>PS binds to (phagocyte / cell surface) receptor (to stimulates a response) ;</p> <p><b>A</b> phagocyte has a binding site for PS / PS binds to a binding site</p> <p><i>ref. to (PS presence / change in membrane) causes chemotaxis / attracts phagocyte (to red blood cell)</i></p> <p><i>treat as neutral suggested chemicals that trigger chemotaxis</i></p> <p>AVP ; e.g. PS acts as an antigen</p>                                                            | 2     |
| 2(d)(ii) | <p>any <b>one</b> from:</p> <p><i>why uninfected red blood cells are destroyed</i></p> <p>(some) PS is present in outer layer so can be, recognised / detected ; AW</p> <p><i>why only some red blood cells are destroyed</i></p> <p>only a few PS in outer layer so lower chance of being recognised ;</p> <p>parts of, parasite / pathogen / <i>Plasmodium</i> (in circulation), may become attached to, surface / cell surface membrane ;</p> <p>(cell) displays, a foreign antigen / a non-self, antigen ;</p> <p>displays an antibody (attached to antigen) ;</p> <p>damaged (by other means) / at end of life-span / non-functioning, (so need destroying) ;</p> | 1     |
| 2(e)(i)  | <p><b>R</b> if context is, red blood cell / entry into red blood cell by pathogen</p> <p>any <b>one</b> from: <i>context is Plasmodium</i></p> <p>for, cell membrane(s) / cell surface membrane ;</p> <p>maintain / regulate, stability / fluidity, of membrane ;</p> <p><b>I</b> affects / changes, <i>if not qualified</i></p> <p><b>A</b> phospholipid bilayer <i>for membrane</i></p> <p>to, convert / AW, to other products needed ;</p> <p>to (breakdown and) use as an energy source ;</p> <p>cannot / does not have gene to, synthesise cholesterol ;</p> <p>AVP ; e.g. to form, vesicle / vacuole (to contain replicated cells)</p>                           | 1     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks    |
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| 2(e)(ii) | <p><b>I</b> ref. to temperature</p> <p>any <b>two</b> from:</p> <p><b>1</b> make more fluid / increase fluidity ;</p> <p><b>2</b> decrease / reduces, membrane stability ;</p> <p><b>3</b> increases, entry / exit, of, polar molecules / ions / water ;</p> <p><b>A</b> increases, permeability / described</p> <p><b>4</b> allow increased (lateral) movement of cell components within bilayer ;</p> <p><b>5</b> reduced interaction with fatty acid tails ;</p> <p><b>6</b> AVP ; e.g. decreased interaction with proteins, so affecting their function<br/>may affect ability to distort shape when passing through capillaries</p> <p><i>if no marks gained, allow one compensation mark for</i></p> <ul style="list-style-type: none"> <li>• changes two of: fluidity / stability / permeability</li> <li>• <b>ecf</b> from 'decreases fluidity'</li> </ul> | <b>2</b> |
| 2(f)(i)  | <p>any <b>one</b> from:</p> <p>active / energy-requiring, process ;</p> <p><b>A</b> need energy, for the process / to move phospholipids<br/>moving, phospholipids / named, against the concentration gradient / to area with higher quantity / AW ;<br/>to allow conformational change (of enzyme) ;</p> <p><b>I</b> active transport but <b>R</b> if described as movement across the membrane</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>1</b> |
| 2(f)(ii) | <p>any <b>one</b> from:</p> <p>actively transport / pump, <math>\text{Ca}^{2+}</math> / calcium ions, out of cell (once entered) ;</p> <p>transports <math>\text{Ca}^{2+}</math> / calcium ions, (out) against the concentration gradient ;</p> <p>cell surface membrane impermeable to, <math>\text{Ca}^{2+}</math> / calcium ions ;</p> <p>no / very few, transport / pump / carrier / channel, (membrane) proteins ;</p> <p><i>context is for entry of calcium ions</i></p>                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b> |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
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| 2(f)(iii) | <p>any <b>three</b> from:</p> <p><b>1</b> scramblase, activated / becomes active / begins functioning / AW ;<br/> <i>context is, increase in <math>\text{Ca}^{2+}</math> / when <math>\text{Ca}^{2+}</math> binds</i><br/> <b>A</b> <math>\text{Ca}^{2+}</math> increases, more scramblase binds to <math>\text{Ca}^{2+}</math><br/> <b>ora</b> scramblase, suppressed / inactive (only), in low <math>\text{Ca}^{2+}</math> concentration</p> <p><b>2</b> scramblase moves, PC from outer to inner layer <b>or</b><br/> PE / PS, from inner to outer layer ;<br/> <b>I</b> scramblase randomly moves, PC / PE / PS</p> <p><b>3</b> described effect on, distribution / organisation / asymmetry / AW (of phospholipids in cell surface membrane) ;<br/> e.g. becomes, disrupted / disorganised / more random<br/> is uncontrolled / not regulated<br/> becomes more, symmetrical / even / evenly balanced<br/> higher proportion of, PC in inner layer<br/> higher proportion of, PE / PS, in outer layer (than normal)<br/> <i>(idea that the phospholipids are in the incorrect location)</i></p> <p><b>4</b> scramblase activity higher than activity of, flippase / floppase<br/> <b>or</b><br/> <i>idea that</i>, flippase / floppase, action cannot counteract effect of scramblase ;</p> <p><b>5</b> ATP runs out, qualified ; e.g. to maintain asymmetry<br/> for (increased), flippase / floppase, activity<br/> to move phospholipids (against their gradient)</p> <p><b>6</b> AVP ; e.g. <math>\text{Ca}^{2+}</math> is, a cofactor / needed for scramblase function</p> | 3     |